IGCSE 0607 Practice Questions — E2.4

Indices II

Total Questions: 12 — **Total Marks:** 66 Marks

Table of Contents & Quick Navigation

- **E2.4** Indices II
  → **Paper 2 (Non-Calculator)** *(10 Qs)* · **Paper 4 (Calculator)** *(2 Qs)*

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1 **(a)** Simplify.

$$4x^2 \times 3x^{-5}$$

..... **[1]**

(b) Simplify.

$$\left(\frac{8x^9}{27}\right)^{\frac{1}{3}}$$

..... **[2]**

2 **(a)** Simplify.

$w^4 \times w^7$

..... **[1]**

(b) Simplify.

$(3x^3)^2$

..... **[2]**

(c) Simplify.

$$\frac{15y^8}{5y^{-2}}$$

..... **[2]**

3 **(a)** Simplify.

$p^4 \times p^7$

..... **[1]**

(b) Simplify.

$(3q^3)^2$

..... **[2]**

(c) Work out the value of $27^{\frac{2}{3}}$.

..... **[2]**

4 **(a)** Work out the exact value of

$$32^{-\frac{2}{5}}$$

..... **[2]**

(b) Solve the equation.

$$5^{x+2} = 125^{x}$$

$x =$ ..... **[3]**

5 **(a)** Simplify fully.

$$4x^{-3} \times \frac{3}{2}x^{\frac{1}{2}}$$

..... **[2]**

(b) Solve the equation.

$$3^{2x+1} = 27$$

$x =$ ..... **[2]**

(c) Simplify fully.

$$\left(\frac{8a^6}{b^3}\right)^{\frac{1}{3}}$$

..... **[2]**

6 **(a)** Simplify fully.

$$\left(\frac{16x^{12}}{y^8}\right)^{\frac{3}{4}}$$

..... **[3]**

(b) Solve the equation.

$$2^{x+3} = 8^x$$

$x =$ ..... **[3]**

7 **(a)** Solve the equation.

$$16^{2x-1} = 8^{x+2}$$

$x =$ ..... **[3]**

(b) Simplify fully.

$$\left(\frac{81x^{12}}{16y^{8}}\right)^{-\frac{3}{4}}$$

..... **[3]**

8 **(a)** Simplify fully.

$$\frac{4x^{-\frac{1}{2}} \times 3x^2}{6x^{\frac{1}{3}}}$$

..... **[3]**

(b) Solve the equation.

$$9^{2x-1} \times 3^{x+1} = 27^x$$

$x =$ ..... **[4]**

9 **(a)** Simplify fully.

$$\frac{12x^{\frac{3}{2}}y^{-2}}{4x^{-\frac{1}{2}}y^{4}}$$

..... **[3]**

(b) Solve the equation.

$$4^{2x-1} \times 8^{x+1} = 32^{x}$$

$x =$ ..... **[4]**

10 **(a)** Simplify.

$$4x^3 \times 5x^{-2}$$

..... **[1]**

(b) Simplify.

$$\left(\frac{2x^4}{y^3}\right)^2$$

..... **[2]**

(c) Solve the equation.

$$3^p = \frac{1}{27}$$

$p =$ ..... **[1]**

11 **(a)** Solve the equation.

$16^x = 2$

$x =$ ..... **[1]**

(b) Work out the exact value of

$125^{\frac{2}{3}} \times 5^{-2}$

..... **[2]**

(c) Solve the equation.

$5^{2x-1} = 125^x$

$x =$ ..... **[3]**

12 **(a)** Simplify fully.

$$\frac{24x^{\frac{3}{2}}y^{-2}}{6x^{-\frac{1}{2}}y}$$

..... **[2]**

(b) Solve the equation.

$$9^x \times 27^{x-1} = 3\sqrt{3}$$

$x =$ ..... **[4]**